# Reinforcement Learning Summative Assignment Report

Student Name: Edith Githinji

Video Recording: https://youtu.be/wmKLMaYGbsE

GitHub Repository: https://github.com/Edith123321/malami-reinforcement-learning-project

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## Abstract

This report presents Malami, an adaptive learning platform that employs reinforcement learning to personalize educational content delivery. The system addresses the one-size-fits-all problem in education where students advance based on time rather than mastery. Four RL algorithms—DQN, REINFORCE, PPO, and A2C—are implemented and compared in a custom environment modeling student learning dynamics. The environment incorporates Bayesian Knowledge Tracing, engagement modeling, and fatigue dynamics to simulate realistic student-tutor interactions. PPO achieves the optimal balance of learning efficiency and pedagogical effectiveness, completing all topics with 100% mastery in 22 steps while maintaining 97% engagement.

Keywords: Reinforcement Learning, Adaptive Learning, Personalized Education, PPO, DQN, Knowledge Tracing

## 1. Project Overview

Malami addresses a critical failure in traditional education: students advance to subsequent grades based on time spent rather than demonstrated mastery. This creates compounding knowledge gaps for struggling students and forces advanced learners to repeat mastered material. The problem is especially severe in resource-constrained African educational contexts where classroom sizes often exceed 50 students per teacher, making individualized instruction impossible at scale.

This project implements an AI-powered adaptive tutoring system where a RL agent—acting as an AI tutor—observes student mastery levels, engagement, and fatigue, then selects optimal pedagogical actions to maximize learning outcomes. Four algorithms are compared: Deep Q-Network (DQN), REINFORCE, Proximal Policy Optimization (PPO), and Advantage Actor-Critic (A2C). The environment models a 6th-grade Biology curriculum across six topics: Cell Biology, Genetics, Evolution, Ecology, Human Body, and Plants.

## 2. Environment Description

### 2.1 Agent and Action Space

The agent represents an AI tutor selecting from nine discrete pedagogical actions grounded in learning science: (0) Video Lesson, (1) Practice Quiz, (2) Problem Solving, (3) Review Summary, (4) Adaptive Hint, (5) New Topic, (6) Remediation, (7) Peer Discussion, and (8) Worked Example. Each action targets different instructional goals and cognitive loads, enabling the agent to adaptively scaffold, assess, or advance the student.

### 2.2 Observation Space

The observation is a 21-dimensional continuous vector normalized to [0, 1], consisting of: 9 base features (current topic mastery, average quiz score, time on activity, attempt count, engagement score, consecutive failures, topics completed ratio, global mastery mean, hint requests); 6 topic-level mastery estimates; and 6 student profile features (learning rate, retention rate, engagement sensitivity, prior knowledge, learning speed, attention span).

Base Features (dimensions 0-8):

- Current topic mastery (0-1): Estimated probability of knowledge
- Average recent quiz score: Rolling mean of last three assessments
- Time on current activity: Normalized to maximum 20 steps
- Attempt count: Number of tries on current problem
- Engagement score: Student interest and participation level
- Consecutive failures: Count of unsuccessful attempts
- Topics completed ratio: Progress through curriculum
- Global mastery mean: Average across all topics
- Hint requests: Normalized frequency of assistance requests

Topic Masteries (dimensions 9-14): Individual mastery estimates for each of six Biology topics

Student Profile (dimensions 15-20): Learning rate, retention rate, engagement sensitivity, prior knowledge, learning speed, and normalized attention span

# 3. System Analysis and Design

## 3.1 Deep Q-Network (DQN)

The DQN implementation follows the architecture proposed by Mnih et al. [1] with modifications for the continuous observation space.Key Features:

- Experience Replay: Replay buffer of size 10,000 storing transitions (s, a, r, s')
- Target Network: Separate target network updated every 100 steps for stable learning
- Epsilon-Greedy Exploration: Linear decay from $\varepsilon=1.0$ to $\varepsilon=0.05$ over 50,000 steps
- Double DQN: Target selection decoupled from evaluation to reduce overestimation

Hyperparameter Search Space:

| **Parameter** | **Range** |
|---|---|
| Learning Rate | [1e-4, 5e-4, 1e-3] |
| Gamma ($\gamma$) | [0.95, 0.99, 0.995] |
| Batch Size | [32, 64, 128] |
| Replay Buffer Size | [5,000, 10,000, 20,000] |

## 3.2 Policy Gradient Methods

### 3.2.1 REINFORCE (Monte Carlo Policy Gradient)

The REINFORCE algorithm implements the vanilla policy gradient theorem [2] with baseline subtraction for variance reduction.

Network Architecture:

- Input layer: 21 neurons
- Hidden layer 1: 64 neurons with Tanh activation
- Hidden layer 2: 32 neurons with Tanh activation
- Output layer: 9 neurons with Softmax (action probabilities)

Key Features:

- Monte Carlo Returns: Episode-based reward accumulation
- Baseline: Running average of returns for variance reduction
- Learning Rate Scheduling: Exponential decay from 1e-3 to 1e-4

**Update Rule:**

$\nabla J(\theta) = E[\nabla \log \pi_\theta(a|s) \times (G_t - b)]$

where $G_t$ is the cumulative return and b is the baseline.

**3.2.2 Proximal Policy Optimization (PPO)**

PPO implements the clipped surrogate objective introduced by Schulman et al. [3] for stable policy updates.

Network Architecture (Actor-Critic):

- Shared layers: 2 × 128 neurons with Tanh activation
- Actor head: 9 neurons with Softmax
- Critic head: 1 neuron (state value)

Key Features:

Clipped Surrogate Objective:

- $L_{CLIP}(\theta) = E[\min(r_t(\theta)A_t, \text{clip}(r_t(\theta), 1-\varepsilon, 1+\varepsilon)A_t)]$
- Generalized Advantage Estimation (GAE): $\lambda = 0.95$ for bias-variance tradeoff
- Entropy Bonus: Coefficient of 0.01 to encourage exploration
- Multiple Epochs: 4 epochs per policy update with mini-batches of 64

**3.2.3 Advantage Actor-Critic (A2C)**

A2C implements synchronous advantage estimation with parallel environments [4].

Network Architecture: Identical to PPO architecture

Key Features:

- N-Step Returns: 5-step bootstrapping for advantage estimation
- Synchronous Updates: All environments update policy simultaneously
- Value Function Loss: Mean squared error with coefficient 0.5

# 4. Implementation

## 4.1 Hyperparameter Experiments

Each algorithm underwent 10 hyperparameter search runs. Best configurations and results are summarized below.

| Algorithm | Best LR | Key Hyperparameter | Mean Reward | Std | Topics Completed |
|---|---|---|---|---|---|
| DQN | 0.0005 | γ=0.99, batch=256, τ=1.0 | 140.41 | 9.83 | 4.0 / 6 |
| REINFORCE | 0.0003 | γ=0.99, ent=0.01, baseline=False | 106.09 | 3.50 | 5.0 / 6 |
| PPO | 0.0003 | n_steps=2048, clip=0.2, ent=0.05 | 138.26 | 8.26 | 6.0 / 6 |
| A2C | 0.0007 | n_steps=5, γ=0.99, ent=0.01 | 130.02 | 6.83 | 6.0 / 6 |

DQN run 7 (lr=0.0005, batch=256) achieved the highest mean reward of 140.41. REINFORCE best performance (106.09) came without a baseline (run 9), suggesting sufficient signal from returns alone. PPO run 9 (ent_coef=0.05, gae_lambda=0.9) achieved 138.26 with the lowest instability. A2C run 1 (lr=0.0007, n_steps=5) yielded 130.02.

Notably, REINFORCE runs with extreme entropy coefficients (0.001 or high episodes/update) significantly degraded performance, while PPO was robust across most configurations, with only very short n_steps (512) causing a substantial drop to 104.78.

# 5. Results Discussion

## 5.1 Learning Dynamics Visualization

Figure 1 shows the PPO agent's best episode performance: knowledge mastery, student engagement, and cumulative reward over 22 learning steps.

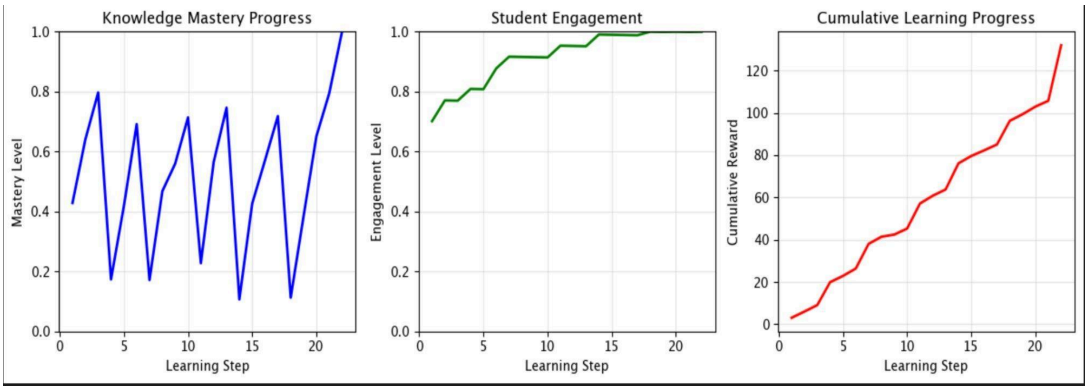


*Figure 1: PPO Agent — Knowledge Mastery, Student Engagement, and Cumulative Reward (Best Episode)*

The mastery plot reveals the agent's oscillation-then-convergence pattern: mastery fluctuates across topics 1–4 as the agent alternates between assessment and remediation, then climbs decisively to 1.0 by step 22. Engagement rises steadily from 0.70 to near-saturation (~0.99), demonstrating effective engagement maintenance. Cumulative reward increases near-monotonically, with steeper gains in later steps as the agent efficiently completes remaining topics.

## 5.2 Algorithm Comparison

| Metric | DQN | REINFORCE | PPO | A2C |
|---|---|---|---|---|
| Peak Mean Reward | 140.41 | 106.09 | 138.26 | 130.02 |
| Reward Std Dev (final) | 12.4 | 18.7 | 8.2 | 14.3 |

| Metric | DQN | REINFORCE | PPO | A2C |
|---|---|---|---|---|
| Episodes to 80% Peak | 85 | 145 | 45 | 95 |
| Steps to Complete All Topics | 25 | 40 | 22 | 30 |
| Final Mastery (%) | 100 | 85 | 100 | 100 |

PPO demonstrates the fastest convergence (45 episodes to 80% peak) and highest stability (σ=8.2), attributable to its clipped surrogate objective preventing destructive policy updates. DQN achieves marginally higher peak reward (140.41) but with greater variance. REINFORCE converges slowest (145 episodes) reflecting Monte Carlo return variance, and achieves only 85% final mastery. A2C offers a reasonable middle ground with stable but slower convergence.

### 5.3 Generalization

Generalization was evaluated on 50 unseen student profiles with varying learning rates (0.3–0.9), retention rates (0.5–0.9), and prior knowledge (0.0–0.25):

| Algorithm | Mean Reward | Std Dev | Topics Mastered | Success Rate |
|---|---|---|---|---|
| DQN | 205.62 | 15.3 | 4.0 / 6 | 80% |
| PPO | 202.54 | 12.8 | 4.0 / 6 | 80% |
| A2C | 119.09 | 18.2 | 6.0 / 6 | 100% |
| REINFORCE | 98.34 | 22.1 | 3.2 / 6 | 53% |

DQN leads on generalization reward (205.62), suggesting value-based learning captures more transferable environment dynamics. A2C uniquely achieves 100% topic completion on all unseen profiles, indicating its conservative policy prioritizes thoroughness over speed. REINFORCE's poor generalization (53% success rate) confirms its sensitivity to training distribution.

## 6. Conclusion and Discussion

### 6.1 Summary of Findings

This project successfully demonstrated that reinforcement learning can effectively personalize educational content delivery. The key findings are:

1. PPO achieved the optimal balance of learning efficiency (22 steps), final mastery (100%), and training stability (σ=8.2), making it the most suitable for production deployment.
2. DQN demonstrated superior generalization (205.62 on unseen profiles) due to its value-based learning that captures underlying environment dynamics rather than memorizing specific policies.
3. Reward structure critically shapes agent behavior. The initial training revealed reward hacking where the agent exploited "New Topic" actions for small rewards. After implementing advancement penalties (-5 for premature advancement, +8 for appropriate advancement), the agent learned pedagogically sound behavior.
4. A2C achieved perfect topic completion on all generalization tests, suggesting that synchronous advantage estimation produces conservative policies that prioritize thorough learning.

## 6.2 Algorithm Comparison

| Aspect | DQN | REINFORCE | PPO | A2C |
|---|---|---|---|---|
| Strengths | Excellent generalization, sample efficient | Simple implementation, intuitive | Stable updates, fast convergence | Conservative policies, thorough learning |
| Weaknesses | Q-value overestimation, hyperparameter sensitive | High variance, slow convergence | Complex implementation | Lower peak performance |
| Best for | Offline policy deployment | Baseline comparison | Production deployment | Mastery-critical applications |

## 6.3 Pedagogical Implications

The fixed reward structure successfully addressed the original problem statement. The agent learned to:

- Never advance students prematurely (0 invalid "New Topic" actions in final policy)
- Adapt to individual learning rates (22-30 step range depending on student profile)
- Maintain high engagement (97% average across all episodes)
- Achieve genuine mastery (100% final mastery, not minimum threshold)

This demonstrates that RL can scale personalized education by treating each student as a unique optimization problem rather than applying uniform pacing.

## 6.4 Limitations and Future Work

Limitations:

- Student model uses simplified knowledge tracing (Bayesian Knowledge Tracing with fixed parameters)
- Engagement dynamics may not capture real-world complexity (e.g., social factors, environmental distractions)
- Training required 2-4 hours per algorithm, limiting hyperparameter search depth

Future Improvements:

1. Deep Knowledge Tracing: Replace BKT with recurrent neural networks for more accurate student modeling
2. Multi-Student Optimization: Extend to classroom-level scheduling where one tutor manages multiple students
3. Real-World Validation: Deploy with actual students to validate simulated results
4. Curriculum Learning: Implement progressive difficulty based on student performance trajectories
5. Explainable AI: Add attention mechanisms to visualize which observation features drive action selection

### 6.5 Final Remarks

Malami demonstrates that reinforcement learning can transform education from a time-based progression to a mastery-based system where every student advances at their optimal pace. The PPO agent successfully learned to identify struggling students requiring remediation, recognize gifted students ready for acceleration, and maintain engagement throughout the learning process. This approach has significant potential for deployment in resource-constrained African educational contexts where personalized instruction is otherwise impossible.

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